



Machine Learning-Based Flood Prediction System

Using Historical Threat Mapping and Live Weather APIs

Amit Bishnoi (1024240005)

Diksha Garg (1024240015)

Introduction & Background

Why today's flood-risk tools fall short



The Limit of Weather Apps

Right now, weather apps show us live data like rain and wind, but they do not tell us if that rain will actually cause a flood in a specific area.



Static Risk Assessment

On the other hand, traditional flood maps are usually drawn once and rarely updated. They cannot react to a sudden, heavy storm.



The Historical Disconnect:

Live weather data and historical disaster records live in separate silos. Current systems struggle to instantly connect a live storm with a location's history of flooding.

METHODOLOGY & PIPELINE

1

API Ingestion

Flask server fetches live temperature, rainfall, and wind speed for the searched location.

2

Spatial Indexing

Haversine formula calculates distance to historical flood centroids, counting past threats within a 100km radius.

3

Feature Fusion

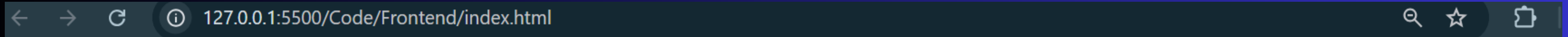
Live weather metrics and historical proximity counts are normalized into a single feature vector.

4

ML Prediction Engine

A Scikit-Learn Random Forest model evaluates the vector to classify the immediate threat level.

Frontend



Flood-Risk AI : Real-Time Prediction Dashboard

Bihar

SEARCH

Live Weather (API)

Temp: 24.5°C

Humidity: 48%

Wind: 4.2 m/s

Rain (Live/Est): 0.0 mm

Historical Threat (DFO)

Past Floods (100km): 1

Avg Past Severity: Low

Recorded Casualties: 0 Recorded

Nearest Flood Epicenter: 142.8 km

ML Risk Prediction

LOW RISK

Atmospheric metrics within normal safety bounds



Back-End Architecture

Core Framework

A python flask rest API orchestrates data ingestion and the ML pipeline

API Integration

OpenWeatherMap API for live atmospheric data and OpenStreetMap Nomination for fallback geocoding

Data Processing

Pandas and numpy handles on-the-fly dataset fusion and Haversine- based mathematical computation

Fail Safe Architecture

Generates localised mock weather data to ensure the ML pipeline never crashes during API rate limits

Use-Cases

Early Warning For Citizens

Search a city during a heavy storm to instantly see if live conditions match past flood trigger.

Emergency Responders

Enables rapid triage by highlighting localised threat zones and historical vulnerabilities on an interactive map.

Urban Planning

Municipalities can identify high-risk intersections of modern weather patterns and past disasters.

Scalable Pipeline

The Exact same logic can be extended to predict wildfires and landslides by swapping the historical dataset.